

Supplementary File 1: Search Criteria for Evaluated Libraries

MechaScreeners: Large Language Model-Based Automated Screening for Systematic Reviews and Research

Introduction

This supplementary file details the exact screening criteria provided to the MechaScreeners Large Language Model (LLM) for each of the 15 reference libraries analysed in the main text. The libraries are divided into 5 development libraries and 10 evaluation libraries (5 focusing on Randomized Controlled Trials [RCTs] and 5 focusing on non-RCTs). Criteria were structured either using the Population, Intervention, Comparator, Outcome, and Type of study (PICOT) format, or provided as general free-text inclusion/exclusion rules. Only the “include” criteria are listed below; fields left blank in the system are omitted.

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1 Development Libraries

1.1 Antibiotic Prescribing and Telehealth

Format: PICOT

Population:

adult or paediatric patients with a history of a community acquired acute infection (respiratory, urinary, or skin and soft tissue)

Intervention:

any type of telehealth consultations (phone or video based)

Comparator:

face to face consultations

Outcome:

the number of antibiotic prescriptions in each type of consultation, follow-up visit rates, testing rates or number of samples sent to the laboratory, any reported adverse events

Type of study:

randomized trials or observational studies

1.2 Long COVID

Format: PICOT

Population:

people of all ages who were eligible to receive a covid-19 vaccine

Intervention:

any dose of a covid-19 vaccine

Comparator:

no vaccination, an active non-covid-19 vaccine control (eg, influenza vaccine), or placebo

Outcome:

patients with a diagnosis of long covid, prevalence of individual symptoms of long covid, such as prolonged fatigue, shortness of breath, cognitive difficulties, and loss of sense of smell

1.3 Natural History of Primary Care

Format: General/Other

Eligibility criteria were: clinical practice guidelines produced by an international or national organisation involved in the publication of guidelines, and contained clinical recommendations on the management of acute infections that are commonly seen in primary care and may be managed with antibiotics (including acute respiratory infections (ARIs), lower urinary tract infections (UTIs), skin and soft tissue infections (SSTIs), conjunctivitis).

Inclusion criteria:

1. Acute respiratory infections including acute otitis media, sore throat, pharyngitis, tonsillitis, common cold, sinusitis, laryngitis, acute cough, acute bronchitis, bronchiolitis middle ear inflammatory conditions
2. Rhinosinusitis, rhinitis, group A streptococcal infections, streptococcal pharyngitis
3. Influenza-like illnesses including seasonal influenza

4. Lower urinary tract infections, cystitis, acute bladder infection, recurrent UTI
5. Skin and soft tissue infections, cellulitis, acne, impetigo, boils, carbuncles, ecthyma
6. All forms of conjunctival conditions
7. Management of croup
8. Urinary tract infections in non-pregnant women
9. Acute exacerbation of COPD

1.4 Non-Drug Interventions

Format: General/Other

Include a study if it meets ALL of the following criteria:

1. **Study Type:** The study must be a systematic review or a scoping review.
2. **Topic:** It must identify the barriers or enablers of using non-drug interventions (e.g., diet, exercise, psychological support, self-management).
3. **Condition:** The intervention must be for the prevention or management of a chronic disease.
4. **Setting:** The setting must be primary care, where a clinician referred the patient to the intervention.
5. **Data Source:** The review must be based on primary studies that are mostly qualitative or mixed-methods.

1.5 Salt Substitution

Format: PICOT

Population:

adults (aged ≥ 18 years)

Intervention:

studies of salt substitution interventions (that is, salt substitute intake compared with regular salt intake) delivered through advice and recommendation from a clinician or researcher, or the provision of salt substitute product

Comparator:

studies that used no intervention or regular table salt as a comparator

Outcome:

mortality, MACE (for example, any composite of myocardial infarction, revascularization, stroke, and heart failure), serious adverse events (for example, renal failure or severe hyperkalemia), SBP and DBP

Type of study:

randomized trials

2 Evaluation Libraries (RCTs)

2.1 Balneotherapy for Chronic Venous Insufficiency

Format: PICOT

Population:

people with venous insufficiency

Intervention:

patients treated with balneotherapy

Outcome:

Edema OR Leg Ulcer OR Pain OR Skin Pigmentation OR Adverse Events OR Quality of Life

Type of study:

randomized trials

2.2 Calcium and Vitamin D for Increasing Bone Mineral Density in Pre-menopausal Women

Format: PICOT

Population:

Adult women

Intervention:

People given Calcium or Vitamin D

Outcome:

Increased bone mineral density or reduction in fractures or quality of life improvements or adverse events

Type of study:

randomized trials

2.3 Methylxanthine for the Prevention and Treatment of Apnea in Preterm Infants

Format: PICOT

Population:

Preterm infants

Intervention:

People given Methylxanthine or Aminophylline or Theophylline or Caffeine

Outcome:

Positive Pressure Therapy or Chronic Lung Disease or Apnea or Death or Neurodevelopmental Disorder

Type of study:

randomized trials

2.4 Phosphodiesterase 5 Inhibitors (PDE5i) for the Treatment of Raynaud's Phenomenon

Format: PICOT

Population:

Adults with Raynaud's Phenomenon

Intervention:

People given Phosphodiesterase 5 Inhibitors

Outcome:

Frequency of attacks or Duration of attacks or Severity of attacks or Raynaud's condition score or Pain or Adverse Events OR Patient Improvement

Type of study:

randomized trials

2.5 Topical and Oral Steroids for Otitis Media with Effusion (OME) in Children

Format: PICOT

Population:

Children with Otitis Media with effusion

Intervention:

People given steroid therapy

Outcome:

Hearing or Quality of Life or Adverse Events or persistence of otitis media

Type of study:

randomized trials

3 Evaluation Libraries (Non-RCTs)

3.1 COVID-19 Measures in Schools

Format: PICOT

Population:

people involved in school settings

Intervention:

measures to reduce transmission of COVID-19

Outcome:

“transmission-related outcomes” OR “healthcare utilisation” OR “economic outcomes” OR “social outcomes” OR “ecological outcomes”

3.2 Diabetes as a TB Risk Factor

Format: PICOT

Population:

patients with diabetes

Outcome:

incidence of tuberculosis

Type of study:

cohort studies

3.3 Falls Interventions

Format: PICOT

Population:

patients over 45 years old

Intervention:

“fall prevention” OR “exercise” OR “safety education” OR “nutrition therapy”

Comparator:

usual care

Outcome:

“incidence of falls” OR “hospitalisations for falls” OR “fall-related injuries” OR “cost of healthcare associated with falls”

Type of study:

“randomised controlled trials” OR “step-wedge designs” OR “non-RCTs”

3.4 Topical Fluoride causing Dental Fluorosis

Format: PICOT

Population:

children exposed to topical fluoride

Intervention:

topical fluoride

Comparator:

“non-topical fluoride treatments” OR “placebo” OR “no intervention”

Outcome:

prevalence of fluorosis

Type of study:

“randomised controlled trials” OR “observational studies” OR “cohort studies” OR “case control studies” OR “cross-sectional surveys”

3.5 Sanitation for Prevention Diarrhoea

Format: PICOT

Population:

“children” OR “adults”

Intervention:

sanitation measures

Comparator:

“open defecation” OR “current practice”

Outcome:

prevalence of diarrhoea

Type of study:

“randomised controlled trials” OR “quasi-RCTs” OR “non-RCTs” OR “controlled before and after studies” OR “matched cohort studies”

4 System Prompt and Prompt Structure

For every reference, MechaScreener sends the LLM a fixed system prompt followed by a user prompt into which the user’s eligibility criteria and the reference’s title and abstract are inserted verbatim. The assistant response is prefilled with the token **Score:** so that the model returns only a single ordinal value. The exact templates used in this study (with the score range set to 1–5) are reproduced below; text in square brackets denotes the fields substituted at runtime.

System prompt:

```
You are a research assistant tasked with evaluating the relevance of
research articles based on their title and abstract. You will score the
relevance on a scale of 1 to 5, where 1 is completely irrelevant and 5
is highly relevant. Consider all aspects of the given criteria when
determining the score. Provide only your score with no explanation.
```

User prompt:

```
The criteria for this research project are:
[USER-PROVIDED ELIGIBILITY CRITERIA]
```

```
Title of the research article:
[TITLE]
```

```
Abstract of the research article:
[ABSTRACT]
```

```
Your response must follow this exact structure: Score:[1-5]
```

Prefilled assistant response: Score:

5 MechaScreener Interface

The MechaScreener tool is provided through the Evidence Review Accelerator (TERA), which requires a free account. To allow readers and reviewers to assess the usability of the system without signing up, Figure 1 shows the MechaScreener interface. Users upload their reference library, enter their eligibility criteria (in PICOT format or as free-text inclusion/exclusion rules), and the tool returns an ordinal relevance score (1–5) for each reference.

6 Distribution of Ordinal Relevance Scores

Table 1 reports the number of references assigned each ordinal relevance score (1–5) by MechaScreener in each of the 10 evaluation libraries. The 1–5 value is used as an ordinal ranking rather than a calibrated probability; the inclusion threshold used throughout the study excludes only references scored 1. The distributions are typically right-skewed toward low scores, consistent with most references in a search being irrelevant to the review question.

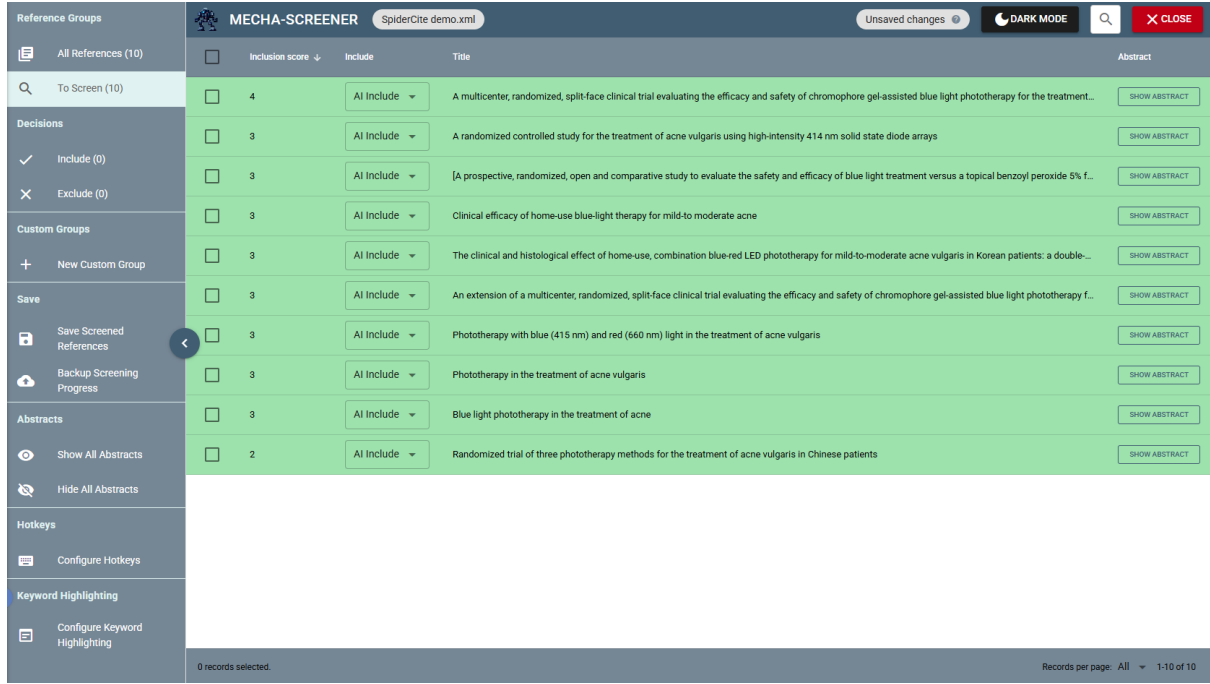


Figure 1: Screenshot of the MechaScreener interface within TERA.

Library Name	Score 1	Score 2	Score 3	Score 4	Score 5
Balneotherapy for Chronic Venous	259	40	14	5	9
Calcium Vitamin D for Bones	1,825	1,779	1,121	480	288
Methylxanthine for Apnea	1,422	386	218	121	299
Phosphodiesterase 5 Inhibitors	2,128	161	19	4	21
Topical and Oral Steroids for Om	3,110	492	43	19	42
COVID-19 Measures in Schools	439	1,275	290	57	80
Diabetes as a TB Risk Factor	1,948	1,600	570	299	594
Falls Interventions	8,102	2,729	2,890	1,849	1,430
Topical Fluoride causing Dental Fluoro- sis	6,398	2,017	737	103	39
Sanitation for Prevention Diarrhoea	8,684	1,067	472	138	112